

Project Overview

- Designed and implemented a **fully functional AMBA AHB-Lite master-slave subsystem** using **Verilog HDL**.
- Demonstrates correct **bus communication, timing, and protocol compliance** for memory-mapped SoC architectures.

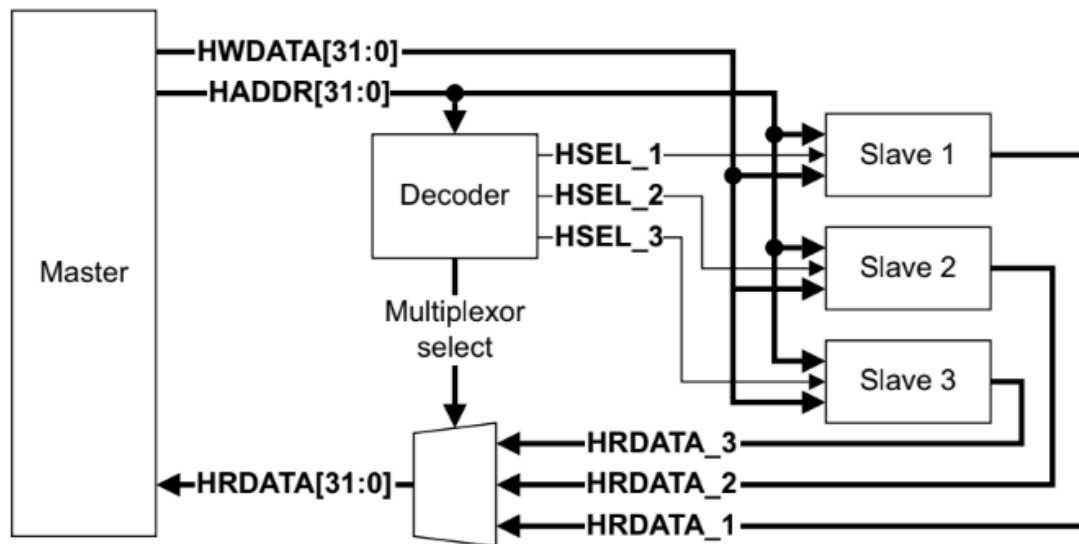


Figure 1-1 AHB-Lite block diagram

Core Components

AHB-Lite Master

- Generates **address, control, and write data** for read/write transfers.
- Uses an **FSM (IDLE → ADDRESS → DATA phases)** to manage AHB-Lite pipeline timing.
- Supports configuration inputs: **address, write enable, burst type, transfer size, and write data**.

- Captures **read data** from slave and signals **transaction completion** (**trans_done**).
- Observes **HREADY/HRESP** from slave for proper handshaking and response control.

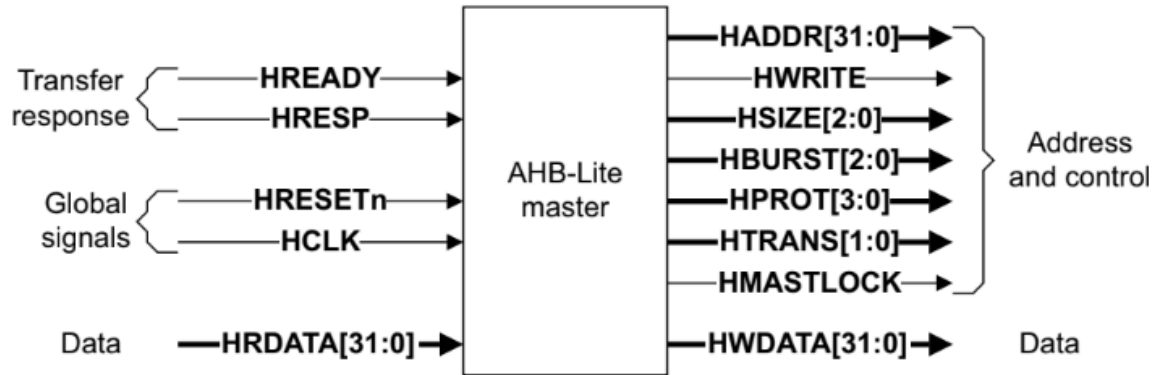


Figure 1-2 Master interface

AHB-Lite Slave

- Implements a **4 KB memory-mapped space** using a **1024 × 32-bit memory array**.
- Decodes captured **address phase information** and performs **byte, halfword, and word accesses**.
- Generates **HRDATA** for reads and **HREADYOUT/HRESP** for valid protocol responses.
- Operates with **zero wait states** (always ready), simplifying initial integration.

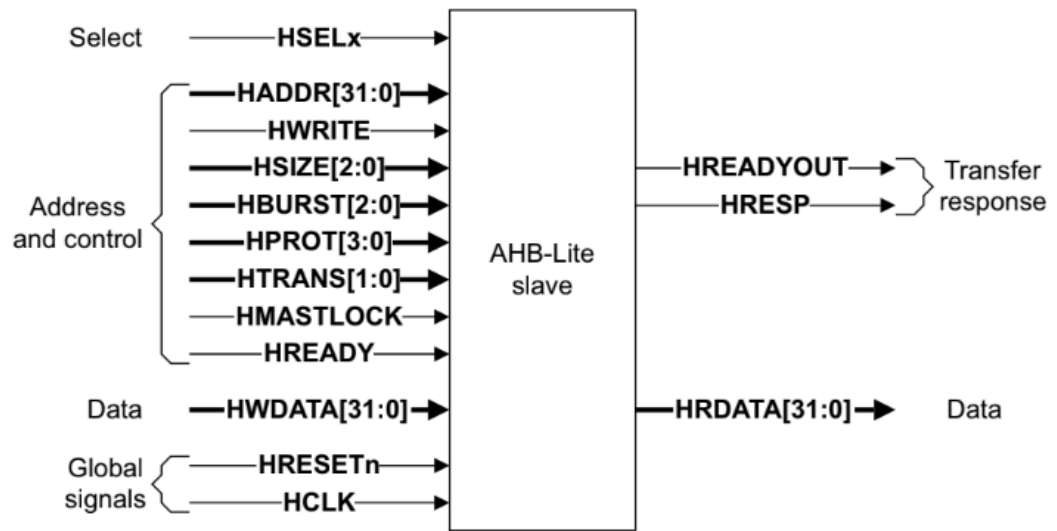


Figure 1-3 Slave interface

AHB-Lite Interconnect

- Provides a **simple 1-to-1 connection** between master and slave.
- Passes address, control, and data signals from master to slave.
- Routes response signals from slave back to master.
- Asserts **slave-select (HSEL)** during active transfers.

Self-Checking Testbench

- Instantiates the **complete AHB-Lite subsystem**.
- Generates clock, reset, and **stimulus for write/read transactions**.
- Supports **word, halfword, and byte** operations for thorough verification.
- Automatically evaluates correctness using **pass/fail tracking**.

- Dumps waveforms (**VCD**) for debugging and visualization.
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Working Flow

1. Testbench triggers a transaction using **start_trans** and provides transfer parameters to the master.
 2. **Master issues address phase signals** (**HADDR**, **HTRANS**, **HWRITE**, **HBURST**, **HSIZE**) followed by **data phase** (write or wait for read).
 3. **Interconnect forwards signals** to the slave and returns response back to master.
 4. **Slave performs memory-mapped read/write** using decoded address and transfer size.
 5. **Master receives HRDATA/HRESP/HREADY**, stores read data if applicable, and asserts **trans_done**.
 6. **Testbench checks output** against expected values and logs pass/fail status.
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Final Outcome

- Achieves a fully verified **end-to-end AMBA AHB-Lite subsystem** with:
 - ✓ Correct single-transfer timing
 - ✓ Memory-mapped read/write functionality
 - ✓ Working testbench with automated validation